Date	25 September 2026
Team ID	SWTID-2026-4307
Project Name	AI Weather Wise API
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4 spaces PEP-8 standard followed
2	Proper File Structure	Files and folders are logically organized	Yes	Modular directory layout (routes, models, utils)
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive Naming (e.g., weather_forecast_data)
4	Function / Method Names	Functions are descriptively named	Yes	Verb-first naming convention used
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings added for core API endpoints
6	Modular Design	Code is split into reusable functions/ modules	Yes	AI model & Weather fetching logic separated
7	No Redundant Code	Duplicate or unused code is removed	Yes	Helper functions used to reduce duplication
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-except blocks with custom HTTP status codes

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Weather Fetcher Module	Python / Requests	Across real-time & forecast API services	High
2	AI Recommendation Engine	Python / Open-AI SDK	In daily advice & activity planner routes	High
3		Python / FastAPI		High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
	Response Formatter Helper		All endpoint JSON output responses	
4	API Token Authenticator	Python / JWT	Protected routes across the API gateway	Medium
5	Cache Manager Module	Python / Redis	Frequent weather query response handler	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean directory structure and well-separated modules.
Readability	5	High readability with self-explanatory names and comments.
Reusability	4	Core modules are highly decoupled and reusable.
Documentation / Comments	4	Thorough docstrings and API documentation provided.
Overall Score	**4.6 / 5**	**Production-ready code meeting high quality standards.**